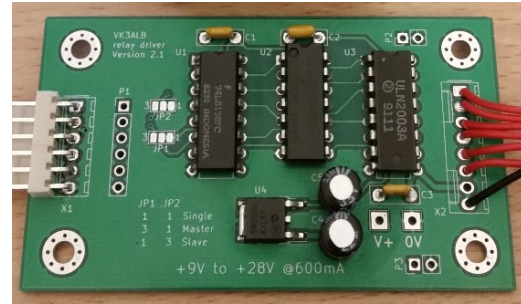


Lou Blasco, VK3ALB
vk3alb@gmail.com
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Introduction

The surplus market often turns up multi-pole coax relays at reasonable prices. These are generally high quality items pulled from test equipment or commercial installations. They usually have very good isolation and can sometimes carry 10's of watts even at 10GHz or beyond.

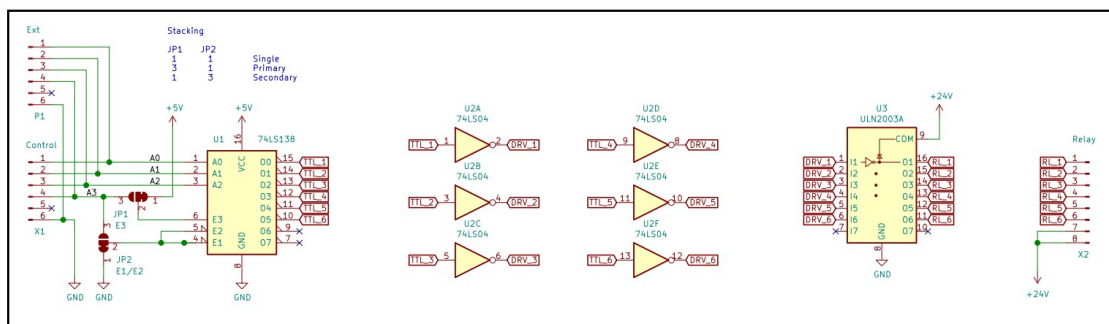
Described, is a simple board that can drive a 6 pole relay with coil voltages ranging from 9V to 28V and can be driven by a micro controller such as Arduino or Raspberry-Pi.

Specifications

- Small 43mm x 75mm PCB
- 3 bit BCD TTL input
- Supports up to 6 way relays
- Supply voltages from 9V to 28V
- Integral 5V supply for TTL IC's
- Uses industry standard ULN2003A open collector driver IC
- Can be stacked to drive two 6 way relays from the same input

Design

The choice of TTL input was driven by the need for micro control. One could argue that a 6PST switch or even 6 lines from a micro could be easier but using BCD saves I/O 3 lines on your micro for other uses.



The circuit is very simple. Starting with a 74LS138 3 to 8 decoder, it reads 3 bit BCD on A0 to A2. The active low outputs drive a 74LS04 HEX inverter through to a bullet proof ULN2003A open collector driver. The TTL supply is derived and regulated by U4. The relay supply is connected to pin 9 of U3 as well as pins 7 & 8 of X2.

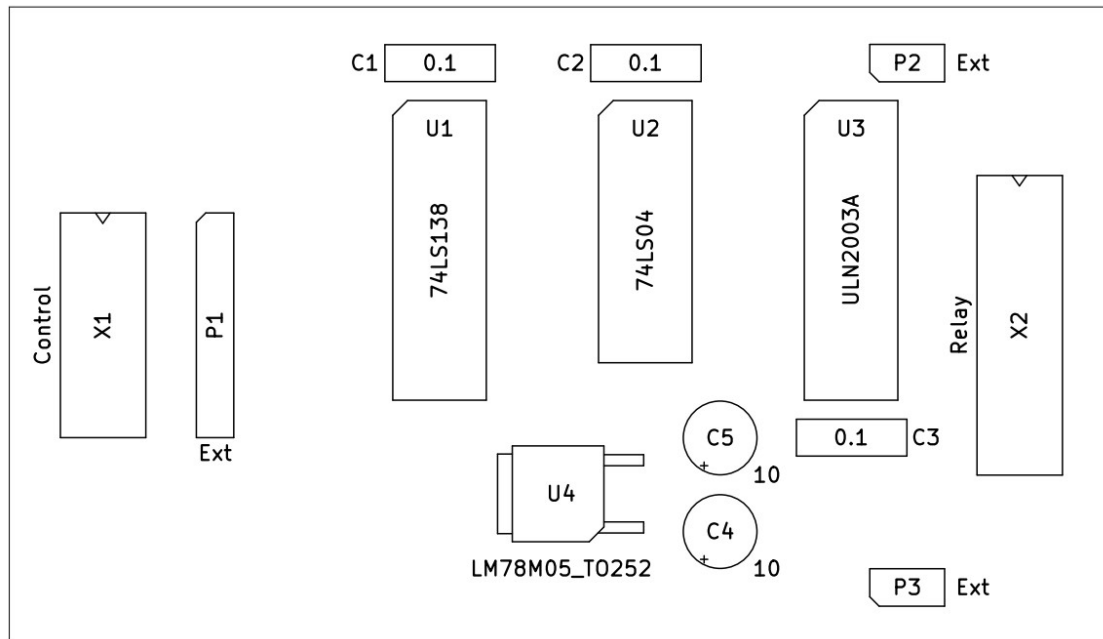
Construction

There's not many part on the board and it shouldn't take too long to assemble. Start by installing 5V regulator U4 ensuring the tab is well sweated to the ground plane. Next fit the IC's and finally the capacitors and whatever connectors you want to use. P1, P2 & P3 should not be installed unless you intend to stack the boards.

When connecting the relay, solder the common line to pin 7 or 8 of X2 and solder the remaining coil wires to X2 in sequence starting from pin 1. Check your work then confirm the board works as expected.

Testing on the prototype indicated the two TTL chip pulled around 15mA. Typical current draw with a single relay coil energized is less than 100mA.

Depending on the relay supply voltage it may be possible to replace the 78M05 with a 78L05 TO-92 package but consider that for 28V the regulator will dissipate ~345mW.



Board layout

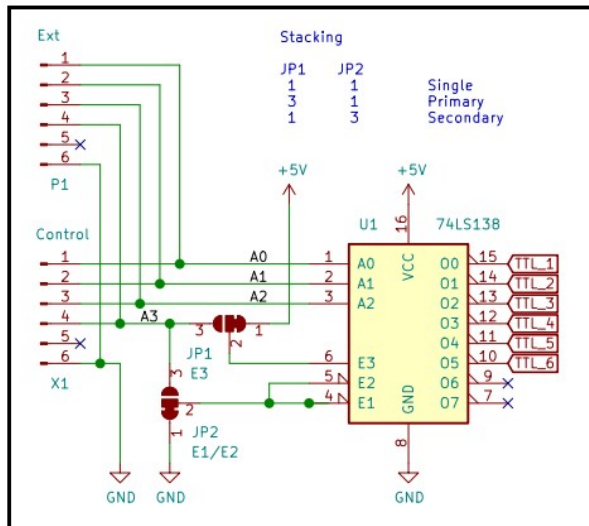
Parts List

U1	74LS138	3 to 8 decoder
U2	74LS04	HEX inverter
U3	ULN2003A	Open collector driver
U4	78M05	Regulator TO-252 package
C1, 2, 3	0.1uf monolithic	
C4, 5	10uf 35V electrolytic	

Miscellaneous connectors, headers, mounting hardware

Stacking

Eagle eyed readers will see JP1 & JP2 which allow two stacked boards to either work in tandem or independently.



JP1	JP2	Mode
1	1	Single
3	1	Primary
1	3	Secondary

To stack the boards, install single row header pins in P1, P2 & P3 on one board and single row header sockets in P1, P2 & P3 on the other then plug the two boards together. Use X1 on either board for the TTL input and only connect the relay supply to one of the two boards as P3 is used to share the supply.

The table shows how the jumpers should be set when stacking boards. Note that the boards are shipped in single configuration. To change the configuration simply cut the tracks between the pads and apply solder bridges accordingly.

Stacking (limitations)

When stacking two boards, pin 4 on X1 becomes the fourth binary bit (A3). It effectively acts as a relay enable pin. One side effect of stacking is that binary 06 and 07 on a single or primary board will disable all relay outputs. Same goes for 0E & 0F for the relay on the secondary board. This may have unintended consequences so take care with your coding. While it is feasible to drive a relay with more than 6 poles using stacked boards, once again, care should be taken with coding.

A3	A2	A1	A0	Relay
0	0	0	0	1
0	0	0	1	2
0	0	1	0	3
0	0	1	1	4
0	1	0	0	5
0	1	0	1	6
0	1	1	0	na
0	1	1	1	na

A3	A2	A1	A0	Relay
1	0	0	0	7
1	0	0	1	8
1	0	1	0	9
1	0	1	1	10
1	1	0	0	11
1	1	0	1	12
1	1	1	0	na
1	1	1	1	na